

3. Consider the Insurances database given below. The primary keys are underlined and the data types are specified.

PERSON (**DRIVER-ID#**: string, name: string, address: string)
CAR (**Regno**: string, model: string, year: int)
ACCIDENT (**report-number**: int, date: date, location: string)
OWNS (**#driver-id**: string, **#Regno**: string)
PARTICIPATED (**#driver-id**: string, **#Regno**: string, **#report-number**: int, **Damage amount**: int)

- i. Create the above tables by property specifying the primary keys and the foreign keys.
- ii. Enter at least five tables for each relation.
- iii. Demonstrate how you
- a. Update the damage amount for the car with a specific Regno in the accident with report number 12 to 25000.
- b. Add a new accident to the database.
- iv. Find the total number of people who owned cars that were involved in accidents in 2002.
- v. Find the total number of accidents in which cars belonging to a specific model were involved

```
CREATE DATABASE InsuranceDB;  
USE InsuranceDB;
```

-- PERSON TABLE

```
CREATE TABLE PERSON (  
    DRIVER_ID VARCHAR(20) PRIMARY KEY,  
    NAME VARCHAR(50),  
    ADDRESS VARCHAR(100)  
);
```

-- CAR TABLE

```
CREATE TABLE CAR (  
    REGNO VARCHAR(20) PRIMARY KEY,  
    MODEL VARCHAR(50),  
    YEAR INT  
);
```

-- ACCIDENT TABLE

```
CREATE TABLE ACCIDENT (  
    REPORT_NUMBER INT PRIMARY KEY,  
    ACC_DATE DATE,  
    LOCATION VARCHAR(100)  
);
```

-- OWNS TABLE

```
CREATE TABLE OWNS (  
    DRIVER_ID VARCHAR(20),  
    REGNO VARCHAR(20),  
    PRIMARY KEY (DRIVER_ID, REGNO),  
    FOREIGN KEY (DRIVER_ID) REFERENCES PERSON(DRIVER_ID),  
    FOREIGN KEY (REGNO) REFERENCES CAR(REGNO)
```

);

-- PARTICIPATED TABLE

```
CREATE TABLE PARTICIPATED (  
    DRIVER_ID VARCHAR(20),  
    REGNO VARCHAR(20),  
    REPORT_NUMBER INT,  
    DAMAGE_AMOUNT INT,  
    PRIMARY KEY (DRIVER_ID, REGNO, REPORT_NUMBER),  
    FOREIGN KEY (DRIVER_ID) REFERENCES PERSON(DRIVER_ID),  
    FOREIGN KEY (REGNO) REFERENCES CAR(REGNO),  
    FOREIGN KEY (REPORT_NUMBER) REFERENCES ACCIDENT(REPORT_NUMBER)  
);
```

-- PERSON

```
INSERT INTO PERSON VALUES  
( 'D1', 'Anil', 'Bangalore'),  
( 'D2', 'Ravi', 'Mysore'),  
( 'D3', 'Sneha', 'Delhi'),  
( 'D4', 'Kiran', 'Mumbai'),  
( 'D5', 'Tina', 'Chennai');
```

-- CAR

```
INSERT INTO CAR VALUES  
( 'KA01', 'Honda City', 2018),  
( 'KA02', 'Hyundai i20', 2020),  
( 'KA03', 'Maruti Swift', 2017),  
( 'KA04', 'Toyota Innova', 2019),  
( 'KA05', 'Tata Nano', 2021);
```

-- ACCIDENT

```
INSERT INTO ACCIDENT VALUES  
(10, '2020-05-01', 'Bangalore'),  
(11, '2020-06-15', 'Mysore'),  
(12, '2020-07-25', 'Delhi'),  
(13, '2020-08-05', 'Chennai'),  
(14, '2021-01-15', 'Hyderabad');
```

-- OWNS

```
INSERT INTO OWNS VALUES  
( 'D1', 'KA01'),  
( 'D2', 'KA02'),  
( 'D3', 'KA03'),  
( 'D4', 'KA04'),
```

```
('D5', 'KA05');
```

-- PARTICIPATED

```
INSERT INTO PARTICIPATED VALUES
```

```
('D1', 'KA01', 10, 5000),
```

```
('D2', 'KA02', 11, 1000),
```

```
('D3', 'KA03', 12, 15000),
```

```
('D4', 'KA04', 13, 2000),
```

```
('D5', 'KA05', 14, 25000);
```

1. Update damage amount for a specific accident

```
UPDATE PARTICIPATED
```

```
SET DAMAGE_AMOUNT = 25000
```

```
WHERE REGNO = 'KA03'
```

```
AND REPORT_NUMBER = 12;
```

2. Add a new accident

```
INSERT INTO ACCIDENT VALUES
```

```
(15, '2026-04-20', 'Pune');
```

3. Find the total number of people who owned cars involved in accidents in 2021

```
SELECT COUNT(DISTINCT P.DRIVER_ID) AS TOTAL_PEOPLE
```

```
FROM PERSON P
```

```
JOIN OWNS O
```

```
ON P.DRIVER_ID = O.DRIVER_ID
```

```
JOIN PARTICIPATED PA
```

```
ON O.REGNO = PA.REGNO
```

```
JOIN ACCIDENT A
```

```
ON PA.REPORT_NUMBER = A.REPORT_NUMBER
```

```
WHERE YEAR(A.ACC_DATE) = 2021;
```

4. Find total number of accidents for cars owned by "Anil"

```
SELECT COUNT(DISTINCT PA.REPORT_NUMBER) AS TOTAL_ACCIDENTS
```

```
FROM CAR C
```

```
JOIN OWNS O
```

```
ON C.REGNO = O.REGNO
```

```
JOIN PERSON P
```

```
ON O.DRIVER_ID = P.DRIVER_ID
```

```
JOIN PARTICIPATED PA
```

ON C.REGNO = PA.REGNO
WHERE P.NAME = 'Anil';

Verify Data

```
SELECT * FROM PERSON;  
SELECT * FROM CAR;  
SELECT * FROM ACCIDENT;  
SELECT * FROM OWNS;  
SELECT * FROM PARTICIPATED;
```

ER DIAGRAM

